Project Initialization and Planning Phase

Date	15 March 2024
Team ID	Individual
Project Title	EV Market Performance & Sustainability Analytics
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Develop an interactive Tableau dashboard to analyse EV sales, vehicle categories, manufacturing countries, battery technologies, driving range, charging and energy-efficiency indicators, and reported annual CO2 savings for evidence-based market analysis.
Scope	The project uses the supplied EV market CSV/Excel dataset in Tableau Desktop. It covers data connection and field checks, KPI cards, sales trends, country sales comparison, vehicle-category and battery-technology distributions, manufacturer comparison, range distribution, a country-category heatmap, a charging-efficiency scatter plot, and dashboard filters where configured. It does not include live feeds or predictive modelling.
Problem Statement	
Description	EV market information includes many dimensions and measures. Comparing sales, prices, range, battery type, manufacturer, geography, and sustainability metrics manually can be time-consuming. A consolidated dashboard is needed to compare these measures consistently and make patterns easier to identify.
Impact	The dashboard can reduce manual comparison effort, make key metrics easier to monitor, and help strategy, business-intelligence, and research users identify areas for further investigation. Conclusions remain limited by the completeness and quality of the supplied dataset.
Proposed Solution	

Approach	Connect the supplied file to Tableau Desktop; inspect field types and missing values; create KPI measures; build sales, country, category, battery, manufacturer, range, heatmap, and scatter-plot views; arrange them into dashboards; and test filters, tooltips, labels, and calculations.
Key Features	Four KPI cards for total sales, average price, average driving range, and annual CO2 saved; sales trend by model year and category; country sales bars; category and battery donut charts; manufacturer comparison; range distribution; country-category heatmap; charging-versus-efficiency scatter plot; interactive filters where configured.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Existing personal computer capable of running Tableau Desktop.	Use the available Windows PC; a dedicated GPU is not required.
Memory	System memory for Tableau and the source dataset.	Recommended 8 GB RAM or more, depending on the available computer.
Storage	Space for the source file and Tableau workbook.	Store the CSV/Excel dataset and .twb/.twbx workbook locally or in approved storage.
Software		
Frameworks	Business-intelligence and dashboard software.	Tableau Desktop; Tableau Public only if the workbook is published there.
Libraries	No separate programming libraries are required for the current Tableau-only implementation.	Not applicable unless additional preprocessing is added later.
Development Environment	Environment used to build, save, and manage the workbook.	Tableau Desktop; Git is optional for version control.
CSV/Excel source connected to Tableau; verify the actual file name, row count, field count, missing values, and units.		
Data	Supplied EV market dataset, described in the project context as approximately 200,000 records and 37 fields.	CSV/Excel source connected to Tableau; verify the actual file name, row count, field count, missing values, and units.